Sustainability Analysis using Power BI

Hands-on Activity

1. Open files one by one to understand what the data these contain.
2. List down the issues that you see.
3. List down the transformations you want to make.
4. List down some things that you want to visualize and present to stakeholders.
5. Import Energy Consumption 2019
6. Explore Power Query Editor
7. Import Building Master and Rates
8. Import data from the folder Other Years folder too.
9. Perform cleanings:
    1. Explore data quality, profile and distribution.
    2. Append Other Years to 2019
    3. Change the Query name to Energy Consumptions
    4. Do not remove the null values yet.
    5. Correct date format
    6. Select the columns Water, Electricity and Gas consumptions
    7. Transform > Unpivot Columns > Unpivot only selected columns
    8. We get 2 columns instead of 3 earlier.
    9. Rename the "attribute" column as **Consumption type**.
    10. Rename the "value" column as **Unit**
    11. The values in the Consumption Type column has repeated " Consumption".
    12. Right click on Consumption Type column > Replace Values > Replace " Consumption" with nothing.
    13. Now the values should be Water, Gas, Electricity.
    14. Clean other queries too.
    15. Go to **Rates** query.
    16. There is no unique column so relationship cannot be created.
    17. Select Year and Energy Type columns.
    18. Go to Add Columns > Merge > Separator none. New column name as "Key". Ok.
    19. New column Key is a combination of Year and Energy Type.
    20. Go to **Energy Consumptions** query.
    21. We need similar key column here too.
    22. We will extract Year from the Date.
    23. Select Date column.
    24. Go to Add Column > Date on the right > Year
    25. It will create a new column called Year.
    26. Now we will merge this new Year column with Consumption Type column.
    27. Select Year column. Control click Consumption Type column.
    28. Go to Add Column > Merge > give new name as Key. Ok.
    29. If you want, can remove the extracted Year column. It serves no purpose.
    30. Explore data quality, profile and distribution.
    31. Check dependencies
    32. If Building Master is not connected to main data, we will do it manually in PBI.
    33. Disable load for Other Years.
    34. Check dependencies again.

    35. Once done, Close and Load
10. See the interface of Power BI Desktop
11. Go to Data View and see. Check the data type icons.
12. Go to Model view and see how the relationships are created for tables.
13. Create relationships if missing for any table.
14. Go to Report View and create any chart of your choice to see how it works.
    1. Click on blank space in Canvas.
    2. From the Data panel, click on the checkbox of Units. It will create a column chart by default.
    3. From the Data panel, click on the checkbox of Consumption Type. It will convert the existing chart to clustered column chart where we can compare the units consumed by each consumption type.
15. Create new measures
16. Go to Model View > Energy Cons. Data
17. New Measure as shown:

```
Total Cost =
SUMX (
    'Energy Consumptions',
    'Energy Consumptions'[Unit] * RELATED ( Rates[Price Per Unit] )
)
```

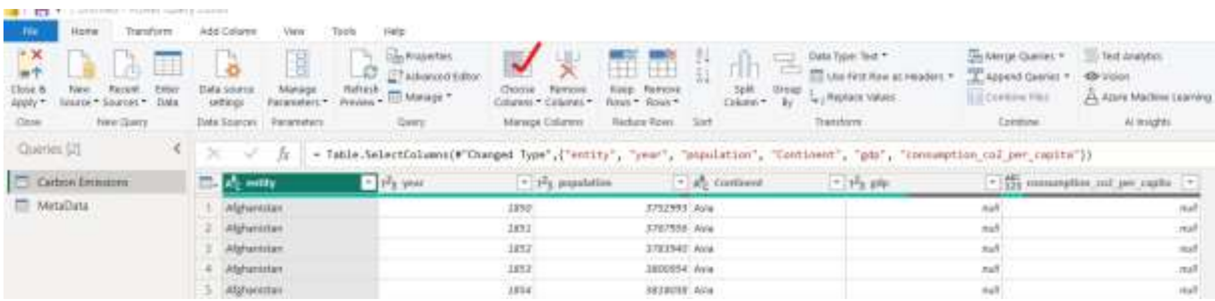
```
Unit Consumed = SUM('Energy Consumptions'[Unit])
```

18. Create different charts as per your choice and what you want to visualize.
19. Try different types of charts using different columns.
20. Select any chart and go to Format to customize its appearance.
21. Do the required formatting.
22. Understand what is Data Storytelling and Stakeholders communications?
23. Go back to Power BI and create the final report in a new page by giving a nice canvas background, proper titles to each chart, removing unnecessary details etc.
24. Right on charts to use features of Summarize and Explain.
25. Best Practices:
    1. Page formatting like templates
    2. Bookmarks
    3. Adding Logo
    4. Good to add NLP features too with a Q&A button.
    5. Publishing to Power BI Service

Case Study 2: **Carbon Emissions**

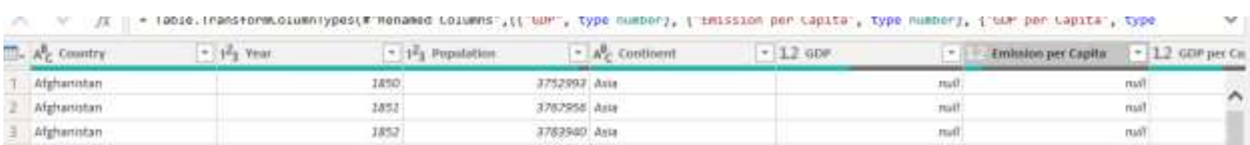
1. Data exploration
    1. Open Carbon Emissions.xlsx file and understand what it contains.
    2. Look at its MetaData worksheet for description of the columns.
    3. There are many columns in the data and many of them have majority of null values.
    4. Some important columns such as GDP, Population which we may want to visualize also contains the null values. This may be because the dataset contains historical data of last 200 years during which data values for some countries may not be available.
    5. We will not remove the ROWS which contains the null value. Instead, we can select that out of all columns, which ones we want to use and then keep those only.

6. Import this data in Power Query Editor.
7. Select both worksheets i.e. Carbon Emissions and MetaData.
8. Select CarbonEmissions query and do the profiling based on entire dataset.
9. Here also we can see many columns with many null values. Since the entity columns, which represents countries, does not contain any null values, we do no need to delete any row.
10. We will just drop out or remove the columns which we do not need.
11. Using this dataset, we are primarily going to focus on:
    i. Top 10 countries by Carbon emissions
    ii. Trend of carbon emissions
    iii. A map visual for Continent and Countries by carbon emissions
    iv. A correlation between two columns among gdp, population and carbon emissions.
    v. A timeline for Consumption-based CO2 emissions per capita vs GDP per capita of countries and continents
12. Go to Home > Choose Columns and select only these columns



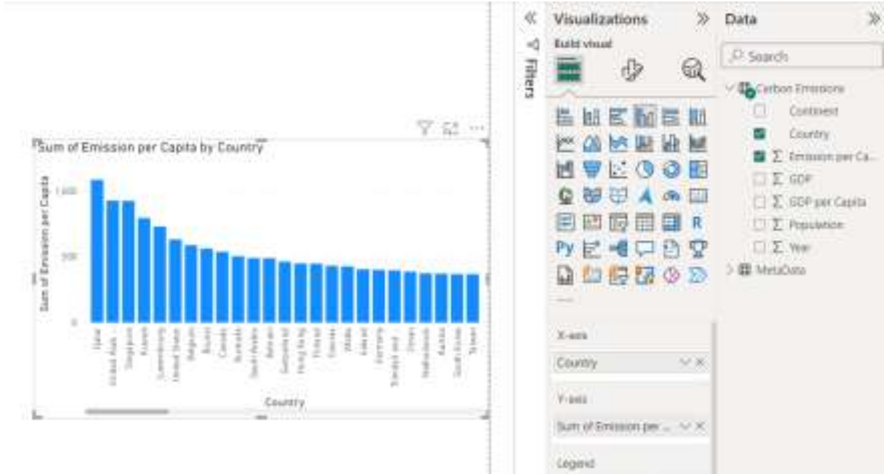
13. For the timeline chart, we will need GDP per Capita too, which is not available in the dataset.
14. Go to Add Column > Custom Column > set new column name as GDP per Capita and use this formula = [gdp] / [population] then click OK. It will create a new column.

15. Ignore the null for now. We will come back to it if it causes any issue.
16. Rename entity as Country, year as Year, population as Population, gdp as GDP, then Emission per Capita. Also correct the data types as shown.



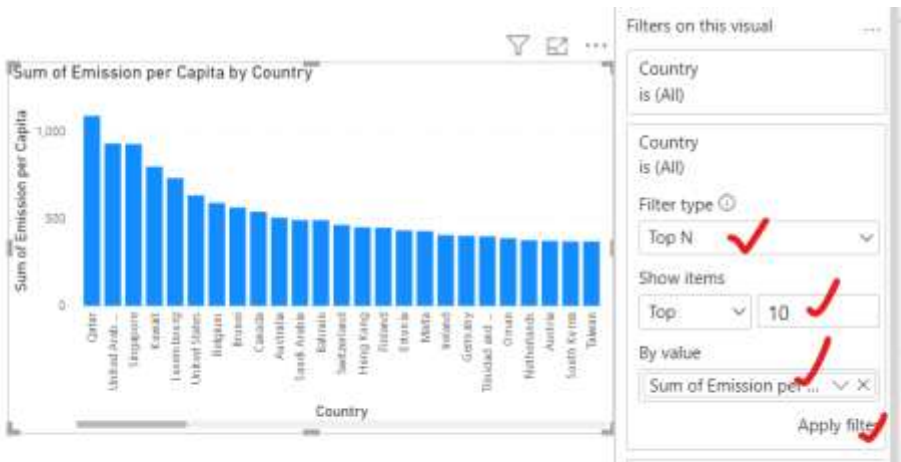
17. Let's not delete the MetaData yet as if required, we can come back here to see the detailed description of other columns and bring them back in our analysis. If you want, can Disable the load for MetaData.
18. Click on File > Close & Apply.

2. Data Visualization
    1. We are going to create the charts now with Carbon Emissions dataset. You can expand it from Data panel to explore what all columns are there.
    2. **Chart:** Top 10 countries by Emission per Capita
        i. Create a chart like this



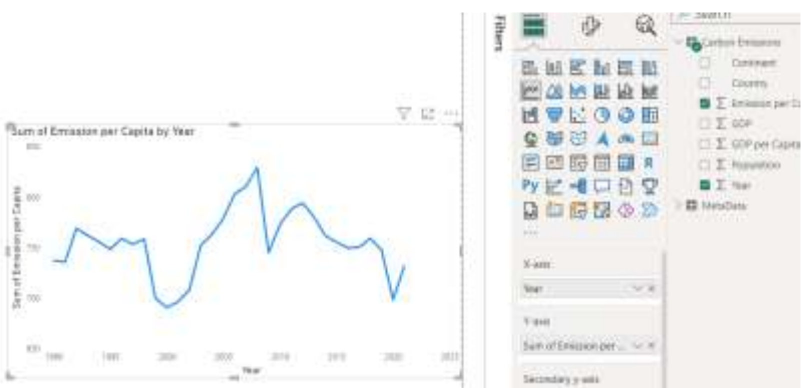
ii. Expand Filter panel. On Filters on this visual > Add data fields here, bring Country.

iii. Apply filter as shown.



3. **Chart:** Trend of Carbon Emissions

   i. Create a chart as shown.

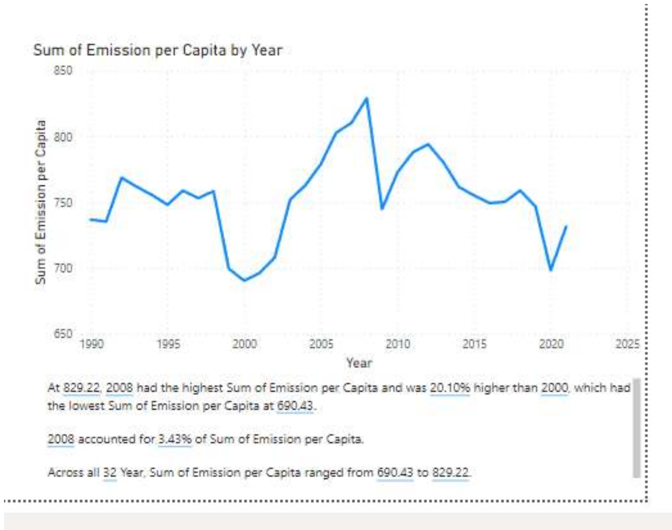


ii. Note there is no hierarchy as on x-axis, we have only Year, not date.

iii. Understand the chart, interpret it.

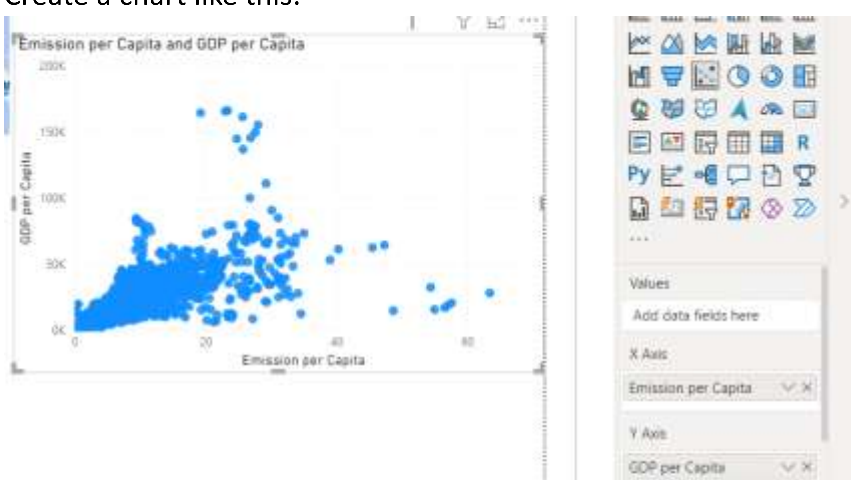
iv. Right click in whitespace in this line chart and select Summarize. Understand the summary. Do you agree with it? If you want to modify this text box, you can do that by just editing inside.

v. Keep this Summary in the page. You can modify this to copy-paste summary from other charts too.



4. **Chart:** Emission by location
    i. Create a chart like this:

5. **Chart:** Correlation between Emission per Capital and GDP per Capita
    i. Create a chart like this:



    ii. You can also add Continent on Legend or can just use the Map visual to highlight for any location.
    iii. For the scatter plot, you can also add a Trend Line from Analytics.
6. **Chart:** Bubble chart with timeline and background image.
    i. Create a chart like this:

   ii. Since there is no data till 1900, you can filter out those years.
   iii. Background image can be added from Format > Plot area background.
   iv. You can also change the Marker type and Size from Format > Visual.

7. **Chart:** Slicers
   i. Add a slicer for Continent and set it to Tile style from Format > Slicer Setting.
   ii. You can add slicer for Year (as slider) and Country (as Dropdown list) too.

3. **Dashboarding**
   1. Now that charts are created, prepare a final page called Dashboard. Give a proper title, image, icons, add charts there to upload on Power BI Service.
   2. Not necessary to keep everything in one page. You can create a new page for the timeline animation.